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Hot streaks and disruptiveness in the career of scientists: is there an association between both phenomena?

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The creative output of scientists may be very different: Some papers may have a disruptive impact on science, some may consolidate existing knowledge, some may be highly influential, and others may yet have untapped potential ("sleeping beauty"). Unlike some previous studies that have explored the relationship between the impact and disruptiveness of papers, this study examines the disruptiveness of research during high impact periods (i.e., "hot streaks") in scientists' careers. By analyzing the career data of 21,271 economists, this study finds that disruptiveness of scientific publications is higher during researchers' hot streaks than in their other periods of careers. Similar to existing works that highlighted a random rule in terms of the occurrence of hot streaks, we also observe the randomness rule for the occurrence of disruptive work among scientists' careers. We find that researchers' most disruptive years are oftentimes earlier than their most highly cited years. A higher publication volume of researchers increases the likelihood of entering hot streaks and producing disruptive publications. We conclude that both scientists and funding agencies can assume the randomness of phases with important papers in scientific careers. But when these phases occur, impactful and disruptive papers can be expected.

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Introduction

Scientific careers are not characterized by constancy. There are periods with more or less publications, and also the importance of the publications varies. The reasons for the dynamics in the careers lie in private (e.g., child care) and professional (e.g., teaching obligations) factors. Research by Liu et al. (2018) reveals that a typical sign of careers are "hot streaks." These are "loosely defined as 'winning begets more winnings' [and] … highlight a specific period during which an individual's performance is substantially better than his or her typical performance" (p. 396). In the science of science context, prior studies define hot streaks as phases during which researchers produce publications with significantly higher citation impact than their typical output (Liu et al., 2018). Many researchers with hot streaks in their careers have one of these exceptional periods; only a few careers are concerned by two or more hot streaks. In an interview, one of the authors of Liu et al. (2018)—Dashun Wang—said that "all of an individual's best works tend to happen within that hot streak" (Mallapaty, 2018).

Some years ago, Wu et al. (2019) proposed the disruption index (D) that can be used to measure the disruptive ($D = 1$) or consolidating ($D = -1$) nature of scientific research. The two types of research can be explained based on the theory of Thomas S. Kuhn (1996) for scientific development: Consolidating science can be denoted as normal science. Normal science is characterized by research that is puzzle-solving within a single paradigm (formed by certain theories, methods, and/or exemplars). Disruptive science can be denoted as scientific revolutions that are characterized by crises in a research field and have paradigm changes as consequences (Wei et al., 2023). D is citation-based and measures the capacity of focal papers (FPs) to overshadow the impact of previously published papers. The disruptive FP does not fit into the consolidating citation network of previous papers but opens new citation paths.

Previous research on the association of citation impact and disruptiveness of research has revealed mixed results: "Intuitively, one would expect disruptive scientific work to be rewarded by high volumes of citations and, symmetrically, impactful work to also be disruptive. A number of recent studies have instead shown that such intuition is often at odds with reality" (Li et al., 2023, p. 1). Li et al. (2023) found that "on average—an author's most disruptive work tends to be well cited, whereas their most cited work does not tend to be disruptive" (p. 1). The mixed previous results on the association of citation impact and disruptiveness of research raise the question about the nature of the connection between citation impact and disruptiveness of research during hot streaks.

This study is intended to clarify the question of the disruptiveness of research during hot streaks: whether hot streaks are not only characterized by high citation impact, but also by disruptive research. In the first step of the analyses, we investigate whether there are differences in disruptiveness before, during, and after hot streaks (measured in terms of citation impact and disruption values) in authors' careers. Since we find these differences in disruption values before and during hot streaks, in the second step of the analyses, we are interested in reasons for the association of hot streaks and disruptiveness. Do greater differences between disruption values before and during hot streaks depend on the career stage of scientists? Does the difference result from the length of hot streaks or the number of publications before and during hot streaks?

Overview of the literature

Review of studies on disruptiveness. The concept of "disruption" probably has different connotations from that of "creativity" or "novelty," although there is no precise definition of disruption (Leibel and Bornmann, 2024). Lee et al. (2015) concluded that creativity is an "evolutionary search process across a combinatorial space and … [they see] creativity as the novel recombination of elements" (p. 685). One popular type of novelty metrics focuses on the atypical combination of prior knowledge based on the notion that creativity is a cumulative process. For instance, researchers assessed the novelty value of papers by analyzing their bibliographies to identify atypical (Uzzi et al., 2013) or unique (Wang et al., 2017) combinations of cited references. Disruption involves not only aspects of novelty but also aspects of impact (Bornmann et al., 2020a; Leibel and Bornmann, 2024). Both Shibayama and Wang (2020) and Leahey et al. (2023) emphasize that only a particular subset of novel research qualifies as disruptive research. The difference between both concepts is reflected in how the indices for measuring the concepts are calculated. D considers both the source and impact of the knowledge (i.e., the cited references and citing papers of FPs) while the novelty indices only consider the source of the knowledge.

D was firstly proposed by Funk and Owen-Smith (2017) as a measure for the magnitude of technology change influenced by a new patent. The authors named the index as CD index. Wu et al. (2019) adopted the index from the technology (measured by patents) to the science context (measured by scientific publications) and named it D. Due to the importance of innovation to both technology and science and the fact that people tend to operationalize technology and science with patents and scientific papers (respectively, that both involve citation relations), Wu et al. (2019) transferred D to bibliometrics to measure the disruptiveness of publications in science. The formula of D for each FP is:

$$D = \frac{N_F - N_B}{N_F + N_B + N_R} \quad (1)$$

where N_F represents the number of publications that cite the FP only without citing any publication that belongs to the references cited by the FP. N_B represents the number of publications that cite both the FP and any one of the references cited by the FP. N_R represents the number of publications that is published later than the FP and cite any one of references cited by the FP without citing the FP itself.

In the following years, researchers mostly from the domain of scientometrics and quantitative science studies started to deal with D and proposed alternatives to overcome identified weaknesses. Leibel and Bornmann (2024) organized the variants of D into five types: variants considering the citation impact (Bornmann et al., 2020b; Funk and Owen-Smith, 2017; Wei et al., 2023), variants dealing with noise (Bornmann et al., 2020a; Deng and Zeng, 2023), variants without N_R (Bu et al., 2021; Shibayama and Wang, 2020; Wu and Yan, 2019), variants disentangling disruption and consolidation (Chen et al., 2021; Leydesdorff et al., 2021), and variants measuring disruption with keywords and Medical Subject Headings (MeSH) terms (Arts et al., 2021; Foster et al., 2015; Hou et al., 2022; Wang et al., 2023). Table 1 shows more detailed information for these five type variants. In recent years, D has not stagnated in its application to specific fields such as urology or radiology. Numerous variants have been proposed, including a variant on the journal level (Jiang and Liu, 2023).

The popularity of studying disruption in scientometrics and quantitative science studies also raises interest in exploring the relationship between disruption and other phenomena in science (careers), e.g., multidisciplinarity (Liu et al., 2024), scientists' productivity (Li et al., 2024) and career age (Cui et al., 2022), diversity of authors' affiliations (Tang et al., 2024), citation inflation (Petersen et al., 2025), novelty/conventionality (Lin et al., 2022), team size (Wu et al., 2019), and the flatness (Xu et al., 2022) and distance (Lin et al., 2023) of scientific

Table 1 Some example variants.

Type	Example variant	Note	Creators
Variants considering the citation impact	$mD = \frac{m_t}{n_t} \times \frac{N_F - N_B}{N_F + N_B + N_R}$	"In this formulation, m_t differs from n_t in that the former counts only citations of the focal patent, whereas the latter includes citations of both the focal patent and its predecessors" (Funk and Owen-Smith, 2017, p. 795)	Funk and Owen-Smith (2017)
Variants dealing with noise	$D_l = \frac{N_F - N_B^l}{N_F + N_B^l + N_R}$	N_B^l represents the number of publications that are published later than the FP and cite at least l references that are also cited by the FP itself	Bornmann et al. (2020a)
Variants without N_R	$D^{noR} = \frac{N_F - N_B}{N_F + N_B}$	D^{noR} removes N_R	Wu and Yan, (2019)
Variants disentangling disruption and consolidation	$D^* = \frac{N_F}{N_F + N_B + N_R}$; $D^{\#} = \frac{N_B}{N_F + N_B + N_R}$	D^* measures disruption, whereas $D^{\#}$ measures consolidation	Leydesdorff et al. (2021)
Variants measuring disruption with keywords and MeSH terms	$ED_R = \frac{n_F - n_{F,R}}{n_F + n_{F,R}}$; $ED_c = \frac{1}{N}\sum_{c=1}^{N} \frac{n_F + n_\varnothing - n_{F,R} - n_R}{n_F + n_\varnothing + n_{F,R} + n_R}$; $ED = \alpha \times ED_R + (1-\alpha) \times ED_c$	ED_R "quantifies the knowledge change directly caused by FP compared to existing research stream" (Wang et al., 2023, p. 154). n_F represents the number of knowledge elements that only exist in FP, whereas $n_{F,R}$ represents the number of knowledge elements that exist in both FP and its cited references ED_c quantifies the average knowledge change from FP's citing papers where N represents the total number of FP's citing papers. For each paper in FP's citing papers, only elements that exist in this paper are counted towards one of n_F, $n_\varnothing$, $n_{F,R}$, and n_R. Here, $n_\varnothing$ represents the number of elements that do not exist in either FP or its cited references ED measures FP's disruption value from the element perspective, where α is a hyperparameter	Wang et al. (2023)

collaboration. In a very influential *Nature* paper, Park et al. (2023) highlight a decreasing trend of disruptiveness for scientific papers and patents since the second world war.

Review of studies on hot streaks. Many studies have dealt with properties of scientific creativity, and the location of the peak time is one of the most favored topics among them. An early study (Lehman, 1958) investigated the 2500 ablest of the world's chemists and found that the maximum production rate is at the age of 34–40. Another study (Simonton, 1991) used four parameters to represent career age, initial creative potential, ideation rate, and elaboration rate. The resulting model of scientific creativity shows an upward trend, a peak, and a downward trend in scientific careers. Such a single-peak phenomenon can be interpreted through the chance-configuration theory (Lehman, 1958; Simonton, 1991). The chance-configuration theory of scientific creativity (Simonton, 1989) emphasizes the crucial role of chance events and structural configurations in scientific innovation. It posits that a scientist's creativity is not solely dependent on personal knowledge and experience but is also influenced by random events and serendipitous discoveries. By effectively configuring and integrating random pieces from previous research, scientists can achieve breakthroughs and innovations.

In this study, we use hot streaks as defined by Liu et al. (2018) to describe success periods in scientists' careers. Hot streaks can be defined as periods when scientists consistently publish papers that subsequently receive more citations than their typical work. Mathematically, Liu et al. (2018) used four parameters to characterize a hot streak: (1) the citation counts of the scientist's publications during the hot streak, (2) the citation counts of the scientist's publications outside of the hot streak, (3) when the hot streak starts, and (4) when the hot streak ends.

Previous studies have investigated several factors contributing to hot streaks. Zhao et al. (2021) report that scientists who publish more papers at a faster rate can have more and longer hot streaks within the same number of years. Li et al. (2020) observed that Nobel laureates (exceptional scientists) are more likely to have more and longer hot streaks in their careers than conventional scientists. Chowdhary et al. (2023) found (for chess players) that beginners experience longer hot streaks than seniors. They also found that "cold streaks" (representing poor performance compared to success) are even longer for beginners. Hot streaks' characteristics depend on disciplines (Győrffy et al., 2020), research type (Weinberg and Galenson, 2019), and the status of scientists' affiliated institutions (Ao et al., 2023).

As the study by Chowdhary et al. (2023) on chess players pointed out, hot streaks are not restricted to science careers. Individuals in arts, social media, and sports also experience periods of intense success. Liu et al. (2018) revealed that hot streaks are common across various careers but manifest uniquely in each field (e.g., painting artists, film directors, and scientists). Oliveira et al. (2023) concluded that music artists' best songs are clustering during the most successful weeks. Seufitelli et al. (2022) compared music artists from different eras and identified similar hot streak patterns across these periods. Individuals in social media (Garimella and West, 2019) have hot streaks as well, where some results from social media may differ from the other fields due to the low barriers for entering social media campaigns (Garimella and West, 2019).

Recent studies have concluded that there is a random rule for the occurrence of hot streaks. Sinatra et al. (2016) pointed out that hot streaks are randomly located in careers. They verified the randomness for scientists who publish in different disciplines, have different career lengths, work in different decades, or publish alone or in groups. Liu et al. (2018) conducted a quantitative study on a large data set and confirmed that hot streaks are randomly located. Although the success of scientists becomes unpredictable due to the random rule, Liu et al. (2021) found that individuals tend to explore different styles or themes in their

scientific work before a hot streak (in an exploration phase), but become more focused after the hot streak (in a consolidation phase). Different from the random rule identified by extant studies, Higashide et al. (2024) note that hot streaks do not exactly obey the random rule and discovered that hot streaks are more frequent in the early or late stages of a scientist's career, both marked by intensive collaboration. The results led them to identify a mid-career pitfall for scientists.

Investigation of disruptiveness during hot streaks. Many previous studies have investigated scientific careers and the prevalence of hot streaks within careers. Hot streaks are usually defined as phases with papers having high citation impact. In light of Dashun Wang's statement about "an individual's best works" (Mallapaty, 2018), we determine whether these best works are solely defined by their high citation impact or if they also encompass other aspects of exceptional research. Since previous studies have shown that research receiving high citation impact must not equal disruptive research, the interplay of disruptiveness and hot streaks on the level of single researcher level becomes relevant. The assumed interplay can be theorized through career mechanisms rooted in cumulative advantage, phases of exploration and exploitation, and paradigm shifts, offering insights into how breakthrough ideas and career trajectories co-evolve.

Hot streaks and disruptiveness may arise synergistically through mechanisms of cumulative advantage, usually encapsulated in the "Matthew effect" (Merton, 1968). Early disruptive contributions—those that introduce novel concepts or methodologies—tend to attract disproportionate attention, citations, and collaboration opportunities. This visibility amplifies a scientist's capacity to sustain productivity, as heightened reputation facilitates access to funding, institutional support, and high-caliber teams. For instance, a disruptive paper that redefines a field may trigger a cascade of follow-up work, enabling the author to enter a hot streak marked by both productivity and continued influence (citation impact).

The interplay between disruptiveness and hot streaks may mirror the tension between exploration (pursuing novel, riskier ideas) and exploitation (refining existing theories and methods). Uzzi et al. (2013) posit that creative breakthroughs often emerge from "atypical combinations" of ideas—a hallmark of disruptive science. However, once a disruptive contribution is made in an exploration phase, scientists go over to exploiting its potential, entering a hot streak characterized by incremental advancements, methodological refinements, or applications. This pattern suggests that disruptiveness may act as a catalyst for hot streaks by opening new research avenues, while the hot streak itself provides the stability needed to consolidate and propagate disruptive insights.

Kuhn's (1996) theory of paradigm shifts may offer a macro-level lens for understanding how disruptiveness and hot streaks intersect. Disruptive contributions destabilize existing paradigms, creating opportunities for scientists to lead new research trajectories. Those who pioneer such shifts may experience prolonged hot streaks as their work becomes foundational to emerging fields. Conversely, the social recognition associated with hot streaks—such as prizes, editorial roles, or leadership positions—can amplify a scientist's ability to promote disruptive ideas, accelerating paradigm shifts. This feedback loop between individual productivity and field-level transformation underscores the socio-cognitive mechanisms possibly linking disruptiveness to career-defining hot streaks.

The assumed co-emergence of disruptiveness and hot streaks in scientific careers may reflect the interplay of cognitive, social, and institutional mechanisms. Disruptive contributions may ignite hot streaks by unlocking new research trajectories, while hot streaks provide the resources and credibility needed to sustain disruptiveness. In order to investigate the co-emergence of disruptiveness and hot streaks, the first phase of the statistical analysis in our study examines whether there are notable differences in disruption values before, during, and after hot streaks, measured in terms of citation impact and disruptiveness of papers, within authors' careers. Upon identifying significant differences in disruptiveness before and during hot streaks, the second phase of our analysis delves into the underlying reasons for the association between hot streaks and disruptiveness. We seek to understand whether the greater differences in disruption values observed before and during hot streaks are influenced by the career stage of the scientists. Additionally, we question whether these differences are a result of the duration of the hot streaks or the number of publications produced before and during these periods.

Methodology

Dataset. In this study, we use the Microsoft Academic Graph (MAG) (Sinha et al., 2015; Wang et al., 2020), a widely-used dataset covering over 200 million publications across various disciplines published between 1800 and 2019. We follow Liu et al. (2018) and only select experienced researchers who have published more than 15 publications in a journal or at a conference and have more than 20 years of career experience. Although we focus in this study on economic[1] researchers to avoid the influence of field-specific differences on the results in the way of doing research, we contrast the results for economics with the results for computer science, geology, and biology (the results for these other disciplines are in the Supplementary Information). There are three reasons for us to focus on economics in the main analyses: First, the mean number of authors of economic publications is relatively small. Each author may have thus a substantial contribution to a publication. Second, economists prefer to use alphabetically sorted authorship lists for publications. Thus, in the analyses of their publications, it is not necessary to consider certain or differing authorship positions. Third, the subfields of economics encompass a wide spectrum including both the social sciences and the sciences. The patterns of disruptive impact may vary between social sciences- and sciences-oriented papers (Park et al., 2023) or theoretical economists and experimental economists (Weinberg and Galenson, 2019). We assume—based on these reasons—that the focus on economics as the primary research subject in this study can make our results more conclusive, ensuring that our findings are stable and widely applicable across different academic backgrounds in economics.

To contrast our results in economics with the results in other fields, we choose researchers in the fields of computer science, geology, and biology as well. We select these fields from a broad spectrum of different disciplines for comparison and generalization of the results in economics. Team size is greater in these fields than in economics and the most contributing author is put in the first position (as in many other fields). We found that the results obtained from computer science, geology, and biology (see Supplementary Information Table S9) are similar to those in economics. The similarity of the results may be interpreted as a hint that our results can be generalized beyond economics.

Characteristics of impact and disruption of publications. We calculate the impact of each publication based on the number of citations. We utilize a variant of D (variants of D are listed in Table 1) to measure the disruptive and developing nature of a publication through its ego-centric citation network. Leibel and Bornmann (2024) summarize three problems faced by the

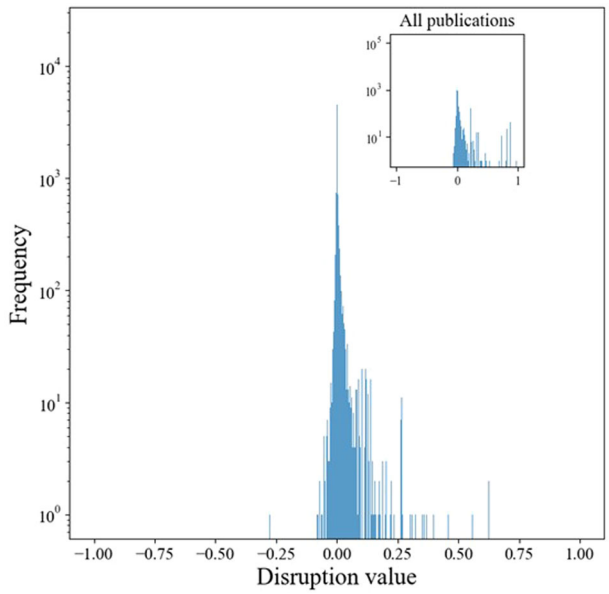


Fig. 1 Distribution of D_5 among the 241,158 publications with five or more citations and five or more cited references. The inset plot shows the D_5 distribution for all 517,093 publications.

calculation of disruption indices: inconsistency, time-sensitive biases, and several data-induced biases. In this study, we target the problems by controlling the citation windows for the FP. First, we limit the citation window for each FP to 5 years, which means the difference in publication year between the FP and each of its citing publications is not greater than 5 years for both disruption and impact measurements. Second, given the high sensitivity of D to the number of references (Ruan et al., 2021) and the sensitivity of the results derived from D to data entries with zero references (Holst et al., 2024), we follow the suggestion by Leibel and Bornmann (2024) to only consider publications with a minimum number of forward citations and references. Here, we use five as the threshold for both sides. All publications, which do not meet the criteria, are thus excluded from further analyses (Wu et al., 2019). The results of robustness checks with a threshold of ten are shown in Table S10 in the Supplementary Information.

Based on the empirical results of previous validation studies, Leibel and Bornmann (2024) recommend to use D_5 instead of the original D in subsequent studies (Table 1). We follow their recommendation in this study and set $I = 5$, where

$$D_5 = \frac{N_F - N_B^5}{N_F + N_B^5 + N_R} \quad (1)$$

N_B^5 represents the number of publications that is published later than the FP and cite the FP and at least five references cited by the FP. Figure 1 shows the distribution of D_5 values among the 241,158 publications in our dataset. The values are mostly small and near zero, which means that there is no distinct difference from the perspective of absolute values. Since one important reason for the small D_5 values is the prevalence of N_R in the index formula, we additionally use D_5^{noR} in this study, where N_R is removed:

$$D_5^{noR} = \frac{N_F - N_B^5}{N_F + N_B^5} \quad (2)$$

We compare the empirical results based on both variants: Whereas the results based on D_5 are in the main text, the results based on D_5^{noR} are in the Supplementary Information. Figure S1 in the Supplementary Information shows the more balanced distribution of D_5^{noR} among the 422,307 publications in the data set compared to D_5 in Fig. 1.

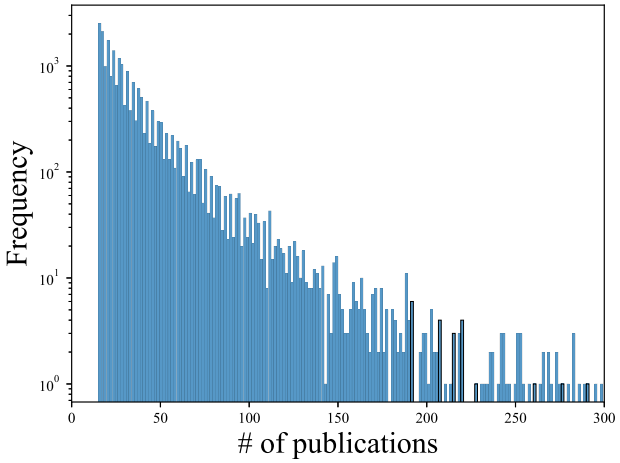


Fig. 2 Distribution of #publications among 21,906 authors.

For impact measurement, we use the citation counts (C_5) for each FP, where

$$C_5 = N_F + N_B \quad (3)$$

Figure 2 shows the distribution of the publications in our dataset among 21,906 authors. The distribution follows the negative power law. When identifying hot streaks, we only consider publications in this study that have citations, because publications with zero citations have (probably) no influence on later publications, at least for the first 5 years considered here (the weighted value of these publications is equal to zero).

Aggregating. In this study, we define the hot streak of an author on the aggregated level: year by year and not on the publication level as done by Liu et al. (2018). We assume that aggregation reduces the contingency of events. Aggregated publications from a certain period are less susceptible to fluctuations caused by atypical or random variations that might be present in single events: One publication reflects a single-time event. That's why we evaluate authors based on periods in their careers instead of focusing on the single publications from their careers.

At least three methods are possible to aggregate the C_5 and D_5 values of all publications from 1 year: mean, sum, and discounted cumulative gain (DCG) (Järvelin and Kekäläinen, 2002). The mean can be influenced by outliers (e.g., a large number of low values) and thus may not accurately represent the overall status of publications[2]. By using the sum of publications as an aggregation method, the overall status of publications may be better represented. The disadvantage of the sum is that it favors scientists with many low-impact papers, rather than rewarding more important publications. The disadvantages of average and sum can be diminished by using the DCG. The DCG has been established in information science for evaluating the quality of retrieval results and its strength. The quality of the retrieval results is assessed by taking into account the position and relevance of documents in the result list. The formula for the DCG is:

$$\mathrm{DCG} = \sum_{i=1}^{n} \frac{rel_i}{\log_2(i+1)} \quad (4)$$

where rel_i represents the relevance score of the i-th result, and $\log_2(i+1)$ reflects the impact of the results' position on the user

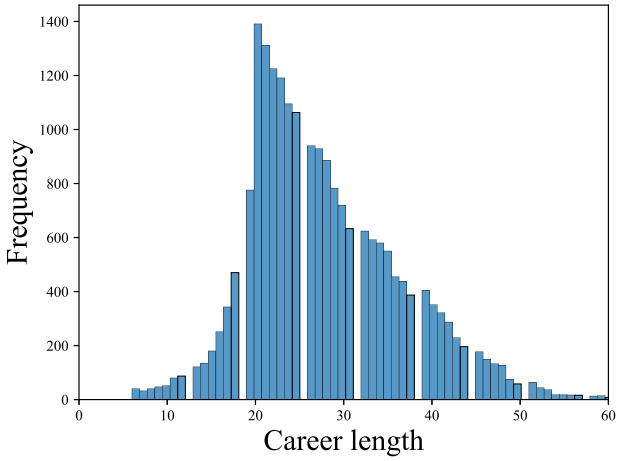


Fig. 3 Distribution of career lengths among 21,271 authors.

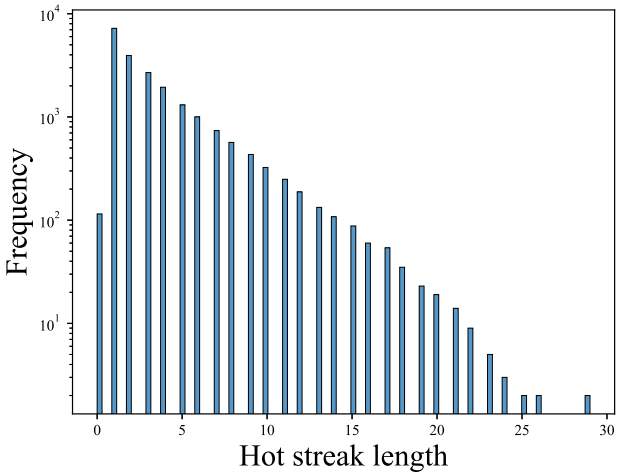


Fig. 4 Distribution of hot streak lengths among 21,271 authors.

experience. DCG exceeds sum- and mean-based aggregation methods by considering the ranking of the disruptiveness or citation impact of publications published by a specific author instead of treating every publication equally.

The DCG has already been used in contexts other than information retrieval: for example, Oliveira et al. (2023) applied it to aggregate the value of songs to an artist. In this study (using bibliometric data), better than mean and sum, DCG is able to consider both value (citation counts or disruption value) and number (number of publications). Although DCG is a variant of the sum method, it includes discounted gains. Here, it is able to reduce the effect of publications with lower citation impact. For example, using the sum or mean methods, two papers with 50 and 5 citations each would be treated similarly to two papers with 25 and 30 citations. DCG instead assigns more weight to the highly cited paper, recognizing its greater impact. DCG may help us thus to better capture the "true" significance of a researcher's work by emphasizing the most influential publications.

DCG is calculated in a two-step process. In the first step, the publications are sorted in descending order of their citation counts or disruption values. In the second step, the DCG is calculated by summing up the weighted values of each publication. The weights are determined by the inverse of the logarithm of the rank (position after sorting) plus one. The formulas for calculating the DCG score for a researcher's publications in terms of citation and disruption values are:

$$\mathrm{DCG}-\mathrm{citation}=\sum_{i=1}^{p}\frac{\log_2(C_i+1)}{\log_2(i+1)} \tag{5}$$

$$\mathrm{DCG}-\mathrm{disruption}=\sum_{i=1}^{p'}\frac{D_i}{\log_2(i+1)} \tag{6}$$

where C_i represents the citation counts of the i-th publication (after sorting), D_i represents the disruption value of the i-th publication (after sorting), p represents the total number of publications, and p' represents the total number of publications with positive disruption values. Instead of the raw number of citations, we take its logarithm due to its skewed distribution. To focus in this study on disruptiveness (and not consolidation), we only consider positive disruption values and exclude negative values when calculating $\mathrm{DCG}-\mathrm{disruption}$[3].

Since publications that have not received any citations have disruption values equal to zero, a disruption value of zero cannot indicate the disruptive or consolidating nature of the work. Thus, we set the start of an author's career at the publication time of the first publication with citations. Once we remove all publications that have not received any citations, some authors, whose career lengths are shorter than 5 years, are removed from our dataset to ensure a sufficient time length for our analyses. This reduces the number of authors in the dataset from 21,906 to 21,271. As Fig. 3 shows, most authors' career lengths are between 20 and 40 years after the removals. In the end, we have two time-series datasets: they condense the career data of a scientist either from the citation impact perspective or the disruptiveness perspective.

Identifying hot streaks. We identify hot streaks for an author as follows: First, we calculate the mean and standard deviation (*std*) of all positive values in a time series dataset. There are some zeros in both datasets (DCG-citation and DCG-disruption values). We set the threshold of the hot streak equal to $mean+\theta*std$. We implement $\theta=0.5$ for subsequent research and conduct robustness tests for other thresholds, e.g., $\theta=0.1,\ 0.2, 0.3, \cdot\cdot, 1.0$. We determine the longest continuous period for each author where the average of DCG-citation values exceeds the threshold. For example, suppose a scientist's career spans 6 years, with the DCG-citation scores of publications each year being 7.08, 9.01, 11.64, 13.85, 5.33 and 12.44. The mean and variance of these scores are 9.89 and 3.02, respectively, resulting in a threshold of 11.40. The years eligible to be the start or end of a hot streak are the third, fourth, and sixth years (as their scores exceed the threshold). The average score of the third and fourth years is 12.75, which exceeds the threshold, and no other longer period has an average score exceeding the threshold (if so, the start year would shift earlier). Therefore, the scientist's hot streak spans from the third year to the fourth year, lasting for a total of two years. As Fig. 4 shows, most authors' hot streak lengths are typically limited to periods no longer than 5 years. Many authors have only one or 2 years of hot streaks. Some authors lack any hot streak; some maintain a prolonged one. We find that the distribution of the hot streak length approximately follows a negative power distribution as shown in Fig. 4.

Regression models. In the regression analyses, we define *D_diff*, the dependent variable, as the difference between the disruption values during and before the hot streak. Our basic assumption is that the disruptiveness before the peak period is less than during the peak period. The independent variables (possibly related variables to the dependent variable) in the analyses are as follows:

(1) Characteristics of hot streaks: As some researchers recorded in our dataset may continue publishing without

ending their scientific career, we adopt *length_hot* and *start_hot_period* as independent variables: *length_hot* represents the length of the hot streak while the latter represents the number of years between the start of a career and the beginning of the hot streak. Since one can assume that longer hot streaks are combined with higher levels of disruptiveness of research, *length_hot* may have a positive effect on *D_diff*. Wu et al. (2022) discovered that scientific innovation is easier for newcomers or individuals from outside the field. The more one learns, the more cognitive burden one carries, making it possibly harder to break through existing knowledge limitations (Jones, 2009). Therefore, the later scientists enter hot streaks (i.e., the larger *start_year_hot*), the less likely they are innovative, resulting in lower disruptiveness (smaller *D_diff*).

(2) Productivity: Liu et al. (2018) found that the success of publications (measured in terms of citations) is random, and the more publications authors publish, the more probable the success will be. We define *#pub_before* and *#pub_during* as the numbers of publications (regardless of with or without citations) before and during hot streaks, respectively. We define *cits_num_before* and *cits_num_during* as the numbers of cited publications (i.e., publications with at least one citation) in the two periods. *D_num_before* and *D_num_during* are the numbers of disruptive publications in the two periods (i.e., those with positive disruption values). Since the period when success occurs in scientific careers has been found to be random (Liu et al., 2018; Sinatra et al., 2016), it has been also shown that the more papers one publishes, the higher the success will be. This may concern not only higher citation impact, but also more disruptiveness. Therefore, *#pub_before*, *cits_num_before*, and *D_num_before*, which reflect scientists' performance before hot streaks, should be negatively correlated with *D_diff*. In contrast, *#pub_during*, *cits_num_during*, and *D_num_during* should be positively correlated with *D_diff*.

(3) Team size: Wu et al. (2019) reveal that large teams are more likely to develop science and small teams are associated with more disruptive research. We take up this result and calculate *team_size_before* and *team_size_during* representing the average team size (i.e., the number of co-authors) of an author's publications before and during the hot streak. We hypothesize that the larger the team size before the hot streak is, the lower the previous disruptiveness, and the larger *D_diff* will be. We also assume that the larger the team size during the hot streak is, the smaller *D_diff* will be.

(4) Time when a hot streak starts: Since Park et al. (2023) show that papers are becoming less disruptive over time, we introduce the start year of the hot streak (*start_year_hot*) as a time variable in our regression analyses. This variable is intended to reduce the impact of different time periods on the results of the regression analyses and to account for cohort effects.

(5) Subfield: Although we focus on statistical analyses in one field (economics), the results may depend on different "cultures" regarding publishing and citing in the corresponding subfields (Bornmann and Wohlrabe, 2019). For example, Weinberg and Galenson (2019) indicate that theoretical economists publish their best work earlier than experimental economists. To consider the effect of subfields, we define *author_field* for a specific author as the subfield where they publish the greatest number of publications.[4] We include the variable as a fixed-effect variable in the regression analyses.

We adopt OLS regressions due to the continuous nature of the dependent variable. The equations for the two models are

$$\begin{aligned} D_diff = {} & \beta 0 + \beta 1 * start_year_hot + \beta 2 * cits_num_before + \beta 3 \\ & * cits_num_during + \beta 4 \\ & * D_num_before + \beta 5 * D_num_during + \beta 6 \\ & * \#pub_before + \beta 7 * \#pub_during + \beta 8 * team_size_during \\ & + \beta 9 * team_size_before + \gamma * author_field \\ & + \epsilon \end{aligned} \tag{7}$$

$$\begin{aligned} D_diff = {} & \beta 0 + \beta 1 * length_hot + \beta 2 * start_hot_period + \beta 3 \\ & * start_year_hot + \beta 4 * cits_num_before + \beta 5 * cits_num_during + \beta 6 \\ & * D_num_before + \beta 7 * D_num_during + \beta 8 * \#pub_before + \beta 9 \\ & * \#pub_during + \beta 10 \\ & * team_size_during + \beta 11 * team_size_before + \gamma * author_field + \epsilon \end{aligned} \tag{8}$$

While Eq. (7) includes only the variables that are not related to hot streaks, together with the start year of hot streaks that is added as a control variable to account for different time periods, Eq. (8) considers all variables.

We conduct five robustness tests to ensure the reliability of our results. First, we test different variants. Besides disruption values, we use the percentile rank of a paper's disruption value within its subfield. Since the results may vary across subfields (Bornmann and Wohlrabe, 2019), we normalize the disruption values by grouping them according to subfields on Level 1. We additionally apply D_5^{noR} as a variant of D_5 to avoid small values around zero. We apply the separating disruption index (Chen et al., 2021; Leydesdorff et al., 2021) of D_5 which separates disruption and consolidation as well as D_1 (Wu et al., 2019). The corresponding results (see Tables S2 and S3 in the Supplementary Information) show that the outcomes in both disruptive and consolidating publications are similar compared to D_5.

Second, we employ several aggregation methods besides DCG, including sum, maximum, mean, and median, to test the robustness of the results based on DCG. The results (see Table S4 in the Supplementary Information) for DCG, sum, and maximum were quite similar, while the mean and median showed some deviations. We believe this is due to the influence of papers with low disruption values. To address this assumption, we modify the mean method: The variant of the mean method calculates the average of values that are greater than or equal to the original mean. Since the median method inherently takes the 50th percentile value, it is also susceptible to the influence of (many) papers with low disruption values. Therefore, we additionally apply the top 10 percentile method, which selects the top 10% of values from the ordered list, arranged from largest to smallest. Similarly, the top 20 percentile method follows the same principle but selects the top 20% of values. The results of the various regression analyses with the alternative aggregation methods reveal that the conclusions remain consistent (see Table S6 in the Supplementary Information). Whereas Table S6 focuses on the results based on D_5, we also produce the results based on D_5^{noR}, see Tables S5 and S7 in the Supplementary Information.

Third, we test different time window lengths before the hot streak, including 1-year, 2-year, 3-year, and a window equal in length to the hot streak itself. The results (see Table S8 in the Supplementary Information) are found to be similar across all time windows.

Fourth, we analyze the papers in various fields besides economics such as computer science, geology, and biology, in order to cover a broad range of disciplines. The results (see Table S9 in the Supplementary Information) show that the conclusions from economics are to some extent generalizable across other fields.

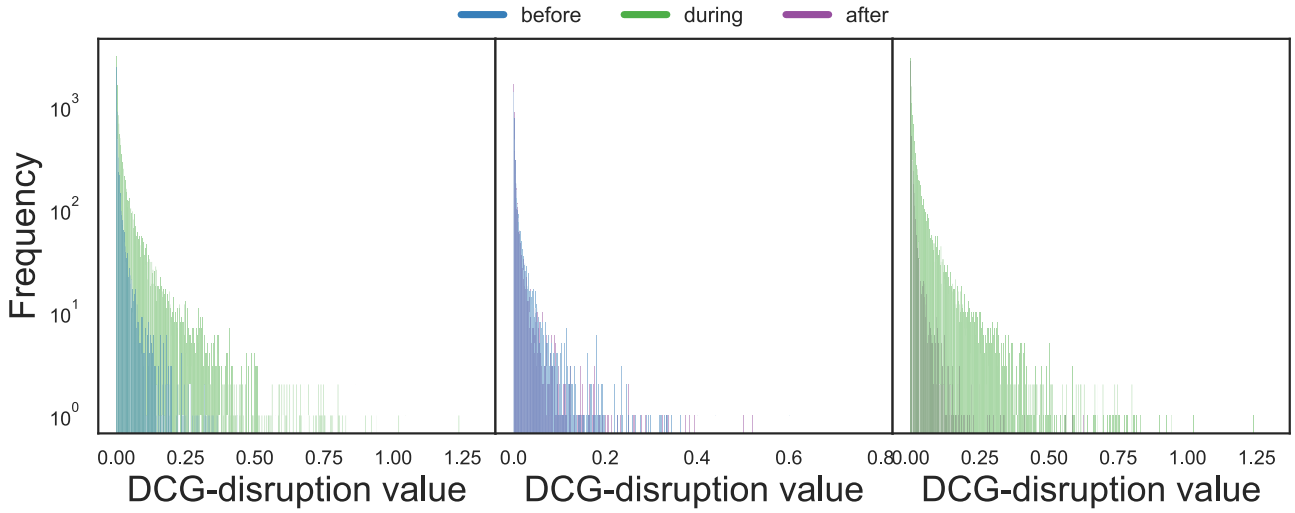


Fig. 5 Distribution of DCG-disruption values in the comparison of two distinct pairs of periods: before and during a hot streak (left), before and after a hot streak (middle), as well as during and after a hot streak (right).

Fifth, while a threshold of five references or five citations was used to exclude publications in the main analysis, we raised the threshold to ten on both sides. The results (see Table S10 in the Supplementary Information) show that the conclusion remains robust with the other threshold.

Results

Descriptive results

Publications during the hot streaks exhibit greater disruption. We investigate whether there are differences in (aggregated) disruption values before, during, and after hot streaks in authors' careers. When measuring disruptiveness before and after hot streaks, a 2-year window is considered, excluding 9.5% of authors whose hot streaks began when their careers started (6.4%) or ended contemporaneously with their careers (3.1%). These considerations reduce the sample from 21,271 to 19,152 authors (the final sample used in the following). The DCG method is employed to aggregate all values for a given period, and we compare the disruptiveness in two different periods based on the distribution of disruption values in Fig. 5. Figure S2 shows DCG-disruption values across different periods in boxen plots. The figures show that disruption values during hot streaks are significantly higher than those before or after, while no significant differences can be observed for the disruption values before and after hot streaks. Using a paired sample *t*-test, we find that the disruption values differed statistically significantly across the three periods. The mean disruption values are highest during the hot streak ($M = 0.03$, $SD = 0.08$), compared to before ($M = 0.00$, $SD = 0.02$) and after the hot streak ($M = 0.00$, $SD = 0.02$). Specifically, the mean disruption value during the hot streak is statistically significantly higher than that before ($t(19151) = 50.87$, $p < 0.0001$) and after ($t(19151) = 51.47$, $p < 0.0001$) the hot streak, and the value before the hot streak is higher than that after ($t(19151) = 4.37$, $p < 0.0001$). Figure S3 shows the results based on D_5^{noR} where the difference is more pronounced than that for D_5. Both results indicate a peak in disruption values during hot streaks. The results thus meet our expectations: Hot streaks seem to be successful periods—not only in terms of citations but also in terms of disruptiveness.[5]

Disruptive research is published earlier than highly cited research. In this section, we focus on the time differences between a scientist's highly disruptive career year and a highly impactful career year. The line plots in Fig. 6 show the average academic year when top *i* most disruptive and most cited years occur during scientists' careers. We see from the figure that the citation curve (blue) decreases monotonically, while the disruption curve (red) first increases and then decreases. A comparison between the two curves indicates that disruptiveness has a tendency to occur earlier. While impact may require long-term accumulation over a scientist's career, disruptiveness can happen earlier in a scientist's career. This is evident from the rapid decline in the mean values (especially those where $i > 2$), which suggests that for the average value to decrease, the newly emerging top years are relatively smaller. Based on these results, we can imagine a scenario where scientists continually engage in disruptive research and, over time, enhance their disruptiveness. During a certain period, their

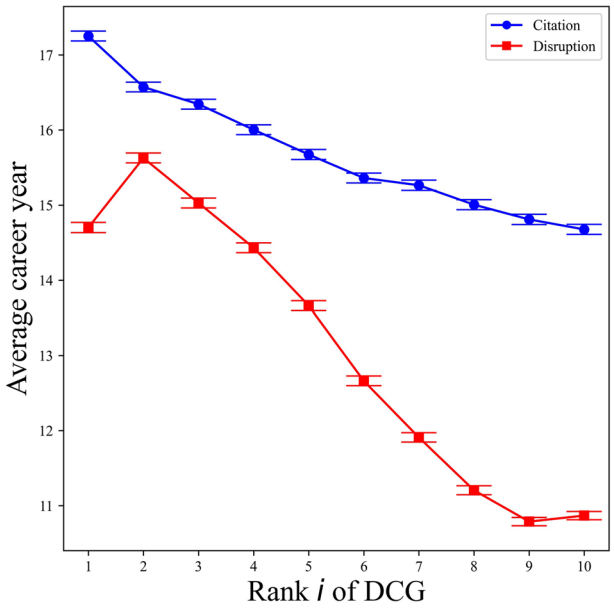


Fig. 6 Average academic years when top *i* most disruptive and cited years occur during scientists' careers. We rank the years based upon the DCG of the sum of disruptiveness/citation count of all publications in that year (the horizontal axis) and examine the average years in scientists' careers. The standard errors are presented as two horizontal bars above and below each average value.

Table 2 Descriptive statistics of variables included in the regression models.

Variable	N	Mean	STD	Min	25%	50%	75%	Max
(1) *D_diff*	19152	0.03	0.08	−0.69	0.00	0.00	0.02	1.30
(2) *length_hot*	19152	3.67	3.33	1.00	1.00	3.00	5.00	29.00
(3) *start_hot_period*	19152	13.09	8.34	1.00	7.00	12.00	18.00	82.00
(4) *start_year_hot*	19152	1.27	9.48	−72.00	−3.00	3.00	8.00	18.00
(5) *cits_num_before*	19152	0.76	0.81	0.00	0.00	0.50	1.00	13.50
(6) cits_num_during	19152	2.41	1.26	0.67	1.67	2.00	3.00	19.80
(7) *D_num_before*	19152	0.22	0.40	0.00	0.00	0.00	0.50	5.50
(8) *D_num_during*	19152	0.88	0.78	0.00	0.33	1.00	1.20	15.00
(9) *#pub_before*	19152	1.97	2.18	0.00	1.00	1.00	3.00	52.00
(10) *#pub_during*	19152	11.54	17.75	1.00	3.00	6.00	13.00	512.00
(11) *team_size_during*	19152	1.53	0.56	1.00	1.00	1.44	1.85	12.00
(12) *team_size_before*	19152	1.19	0.83	0.00	1.00	1.00	1.67	12.00

STD standard deviation, *Min* minimum, *25%* percentile 25%, *50%* median, *75%* percentile 75%, *Max* maximum.

Table 3 Spearman rank correlations of variables included in the regression models.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) *D_diff*	1.00	0.41	0.11	−0.02	0.12	0.23	0.03	0.60	0.10	0.40	0.11	0.05
(2) *length_hot*	0.41	1.00	0.26	0.21	0.37	0.28	0.29	0.20	0.34	0.91	0.20	0.23
(3) *start_hot_period*	0.11	0.26	1.00	0.41	−0.09	−0.07	0.04	0.06	−0.04	0.18	−0.03	−0.06
(4) *start_year_hot*	−0.02	0.21	0.41	1.00	0.03	0.12	0.13	0.16	0.00	0.21	−0.04	−0.02
(5) *cits_num_before*	0.12	0.37	−0.09	0.03	1.00	0.38	0.51	0.20	0.85	0.44	0.15	0.57
(6) *cits_num_during*	0.23	0.28	−0.07	0.12	0.38	1.00	0.29	0.41	0.34	0.60	0.16	0.23
(7) *D_num_before*	0.03	0.29	0.04	0.13	0.51	0.29	1.00	0.32	0.43	0.33	0.14	0.34
(8) *D_num_during*	0.60	0.20	0.06	0.16	0.20	0.41	0.32	1.00	0.14	0.30	0.12	0.10
(9) *#pub_before*	0.10	0.34	−0.04	0.00	0.85	0.34	0.43	0.14	1.00	0.42	0.10	0.58
(10) *#pub_during*	0.40	0.91	0.18	0.21	0.44	0.60	0.33	0.30	0.42	1.00	0.21	0.28
(11) *team_size_during*	0.11	0.20	−0.03	−0.04	0.15	0.16	0.14	0.12	0.10	0.21	1.00	0.36
(12) *team_size_before*	0.05	0.23	−0.06	−0.02	0.57	0.23	0.34	0.10	0.58	0.28	0.36	1.00

works (being disruptive or not over time) reflect a concentrated peak in disruption. After their most disruptive year, they also reaches the most impactful year of their career.

Results of regression analyses

A higher publication volume increases the likelihood of entering hot streaks and producing disruptive publications. Table 2 shows descriptive statistics of the variables included in the regression models. The mean value of *D_diff* is 0.03, with a standard deviation of 0.08, indicating considerable variation in disruptive differences within the sample. *length_hot*, representing the duration of hot streaks, has a mean value of 3.67 years and a standard deviation of 3.33 years. The range is from 1 year to 29 years, showing significant differences in the duration of hot streaks but with a concentration within 7 years (mean + standard deviation). *start_hot_period* represents the starting time of the hot streak, with a range from 1 year to 82 years[6] and a mean value of 13.09 years. This reflects that hot streaks begin at different points in a career, averaging around 13 years after the start of the career.

The mean values of *cits_num_before* and *cits_num_during* are 0.76 and 2.41, respectively, indicating a significant increase in the number of cited papers during hot streaks. *D_num_before* and *D_num_during* show that the number of disruptive papers increases from 0.22 to 0.88, indicating a substantial rise in disruptive papers during hot streaks.

The correlations of variables included in the regression models are shown in Table 3. *D_diff* has a positive relationship with the number of publications during the hot streak. This result is expectable because the DCG method inherently determines that more publications during the hot streak are associated with greater *D_diff*. The positive relationship of *D_diff* and the number of publications before the hot streak is against our intuition. One would expect that more publications before the hot streak are associated with greater disruption values before and resulting in smaller *D_diff* values later. Active publishing (being research active) before hot streaks does not only seem to trigger the appearance of hot streaks but also to increase the general chance of disruptive papers (later on). Inspired by the fact that the probability of success is related to the number of publications (Liu et al., 2018), we speculate that a higher publication volume increases the likelihood of entering hot streaks.

The variable *#pub_during* has a strong relationship with *length_hot*: frequent publishing increases the chance of ongoing hot streaks. This result confirms previous results by Liu et al. (2018). The relation between *cits_num_during* and *length_hot* is weaker than the relation between *cits_num_before* and *length_hot*. It seems that hot streaks profit from higher quality research (measured in terms of citations) before hot streaks begin.

We check the variance inflation factor (VIF) values of the variables included in the regression model. The VIF measures the degree of multicollinearity of one independent variable with the other independent variables as one requirement of the regression analyses. The VIF of the variables is shown in Table 4. The largest VIF is 8.91. Vittinghoff et al. (2012) recommends to use the value 10 as a threshold for the existence of multicollinearity.

Longer hot streaks are more likely to produce highly disruptive works. The results of the regression analyses are shown in Table 5. Because the absolute values of D_5 are very small and concentrated near zero, the variable coefficients in the first two models are extremely small. These small and concentrated regression coefficients cannot be resolved by using a sum-based

aggregation method (Model 5 in Table 5). The same is true for applying standardization (Table S1 in the Supplementary Information). Standardization means transforming all variables into variables with zero mean and unit variance. Since these results are in a lack of clear differentiation, subsequent coefficient comparisons analyses under the same scale are inconvenient. According to Ruan et al. (2021), the number of references may be also related to disruptivity. Thus we include two additional models by adding the number of references as a control variable. These results are shown in Table S11 that are similar to the results in Table 5.

When we adopt the percentile D_5-based model (Model 3 in Table 5), we observe that the differences between coefficients become more pronounced without altering their relative magnitudes. For example, the ratio of coefficients between *length_hot* and *start_year_hot* changes from approximately -3 times (0.003/0.001 in Model 2) to -16 times (18.428/1.153 in Model 3). However, since percentile D_5 represents the relative magnitude of disruptiveness, which is not conducive to a more intuitive analysis, we shift to use D_5^{noR}. D_5^{noR} is a variant of D with a more balanced numerical value distribution that can better reflect the degree of disruptiveness. In Model 4, where the dependent variable is D_5^{noR}, the regression analysis reveals a positive and statistically highly significant coefficient for the independent variable *length_hot* ($\beta = 0.227$, standard error = 0.019, $p < 0.001$). This indicates that longer hot streaks are associated with greater disruption values during these periods. For each additional unit (year) increase in *length_hot*, the D_5^{noR} score increases by 0.227 units, suggesting that the longer hot streaks are, the greater the disruption values during the hot streak.

The variable *start_hot_period* is statistically significant with positive coefficients. This result may be interpreted in the way that the disruptiveness of research does not depend on the career stage. The result corresponds to the random success in scientists' careers from the perspective of citation impact (Liu et al., 2018). The variable *start_year_hot* has a statistically significant but negative effect on the dependent variable. This may indicate the following: while Park et al. (2023) indicates a decreasing trend of publications' disruptiveness over time, our results reveal that the impact of time is not substantial at the author level. Both *start_hot_period* and *start_year_hot* exhibit statistically significant but relatively small[7] and potentially negative coefficients in total. This suggests that the start year of the hot streak has a minor negative influence on the disruption difference before and during the hot streak.

Table 4 VIFs of variables included in the regression models.

Variable	VIF
length_hot	7.47
start_hot_period	3.39
start_year_hot	1.17
cits_num_before	8.57
cits_num_during	8.91
D_num_before	2.30
D_num_during	3.52
#pub_before	6.64
#pub_during	7.33
team_size_during	6.78
team_size_before	4.11

VIF variance inflation factor.

Table 5 Results of regression analyses including influencing factors on the difference in disruption values before and within hot streaks.

	(1) D_5 DCG	(2) D_5 DCG	(3) Percentile D_5 DCG	(4) D_5^{noR} DCG	(5) D_5 SUM
const	0.015***	−0.001	−29.409***	−0.311***	−0.001
	(0.004)	(0.003)	(6.362)	(0.081)	(0.005)
length_hot		0.003***	18.428***	0.227***	0.002
		(0.001)	(1.253)	(0.019)	(0.002)
start_hot_period		0.000***	0.350***	0.004***	0.001***
		(0.000)	(0.050)	(0.001)	(0.000)
start_year_hot	−0.001***	−0.001***	−1.153***	−0.006***	−0.001***
	(0.000)	(0.000)	(0.075)	(0.001)	(0.000)
cits_num_before	−0.001	−0.001	5.841**	0.053	−0.000
	(0.002)	(0.002)	(2.518)	(0.038)	(0.003)
cits_num_during	−0.011***	−0.008***	−12.276***	−0.126***	−0.014***
	(0.001)	(0.001)	(2.230)	(0.032)	(0.002)
D_num_before	−0.015***	−0.016***	−59.355***	−0.955***	−0.008
	(0.003)	(0.003)	(2.112)	(0.043)	(0.005)
D_num_during	0.029***	0.029***	85.637***	1.162***	0.040***
	(0.002)	(0.002)	(3.779)	(0.043)	(0.003)
#pub_before	−0.001	−0.000	−2.902***	−0.031**	−0.002
	(0.001)	(0.001)	(0.857)	(0.013)	(0.001)
#pub_during	0.002***	0.001***	2.026***	0.029***	0.003***
	(0.000)	(0.000)	(0.384)	(0.005)	(0.001)
team_size_before	0.000	−0.001	−2.755***	−0.017	−0.001
	(0.001)	(0.001)	(0.713)	(0.011)	(0.001)
team_size_during	−0.001	−0.001	1.800*	−0.049***	−0.001
	(0.007)	(0.007)	(0.004)	(0.785)	(0.396)
N	19,152	19,152	19,152	19,152	19,152
Fixed effect	Y	Y	Y	Y	Y
R-square	0.189	0.195	0.782	0.822	0.281
Adj. *R*-square	0.188	0.194	0.782	0.822	0.281

***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.

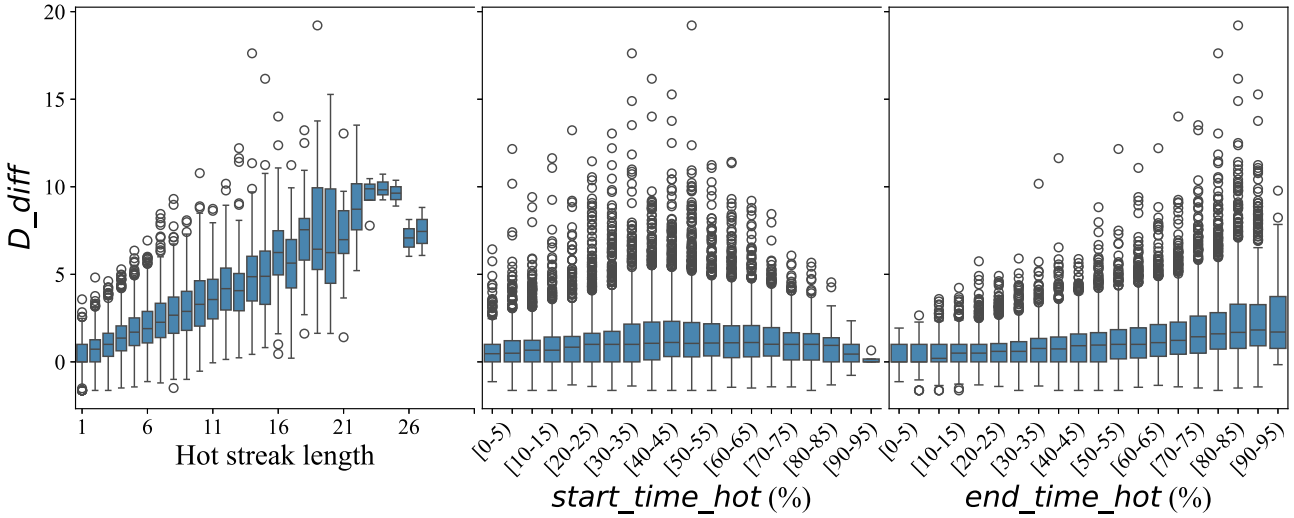


Fig. 7 Relationship between *D_diff* based on D_5^{noR} and hot streak length (left), *start_time_hot* (middle), and *end_time_hot* (right), respectively. Each box in the three panels represents the association between *D_diff* and a given group (defined by the length, start time, and end time of the hot streak). The results that are based on D_5 are shown in Figure S4 in the Supplementary Information.

To better understand the association between disruptiveness and hot streaks, we use three variables (*length_hot*, *start_time_hot*, and *end_time_hot*) to explain the differences in disruption values using the characteristics of hot streaks. *start_time_hot* measures the start time of the hot streak as a proportion of the overall career length, and *end_time_hot* captures the end time as a proportion of the overall career length. As Fig. 7 shows, there is a positive relationship between the hot streak length and the difference in disruption values between during and before hot streaks, which is consistent with the regression results. We mark the publications if their citations are in the top 5% of their field, and explore the relationship between hot streak length and number of top cited publications. The results (see Fig. S5 in the Supplementary Information) point to a strong positive relationship between both variables.

The results in Fig. 7 do not reveal a significant relationship between *start_time_hot*, *end_time_hot* on one side and *D_diff* on the other side. Liu et al. (2018) found that successful periods in careers happen randomly. This randomness may be the explanation for the results with the variables *start_time_hot* and *end_time_hot* in this study.

The number of cited publications negatively affects the overall disruption. The fourth model in Table 5 shows mixed results about the influence of the number of publications. First, *#pub_before* shows inconsistent relations with the dependent variables under different settings (positive and statistically significant, as well as negative and statistically not significant results). Second, *cits_num_before* has a statistically not significant coefficient, while *cits_num_during* has a negative coefficient. These results may point to the conclusion that publishing papers with considerable citation impact during a hot streak lowers the level of disruptiveness.

Third, *D_num_before* and *D_num_during* have negative and positive coefficients, respectively. These results are consistent with our intuition: The more disruptive publications a scientist has during a certain period, the greater the disruptiveness in this period compared to other periods in the career. This conclusion may match the general finding from Liu et al. (2018) that the more a scientist has published in a certain period (papers with more and less citations), the greater the probability of success compared to other periods (measured, e.g., in terms of citations). The models using the maximum to aggregate disruption values (see Table S4 in the Supplementary Information) infer that scientists' ability for highly disruptive research (measured in terms of the maximum disruption value) will improve if they are able to produce (a stream of) disruptive papers ($D > 0$).

Based on the results of the regression analyses in Table 5, it is possible to explore various (fictitious) scenarios for the included variables. For example, a scientist publishes a paper with certain characteristics before or during the hot streak: What is the effect of this paper on *D_diff*? Depending on the characteristics, the paper will probably influence other variables concerning disruptiveness and citations in a specific way. If the paper was published before the hot streak, received some citations, but is not a disruptive paper, the coefficient of that publication is $(-0.031) * 1 + 0.053 * 1 + (-0.955) * 0 = 0.022$ based on the fourth model in Table 5. This means that this publication increases *D_diff* of a scientist by 0.022. Other (fictitious) scenarios are listed in Table 6. For a fictitous publication without citations, the coefficient is negative prior to the hot streak but becomes positive during the hot streak. This pattern is consistent with the previous finding that a higher publication volume increases the likelihood of having disruptive publications. If the fictitous publication is a disruptive paper, its coefficient is negative before the hot streak but positive during the hot streak (this is consistent with our intuition). However, if the publication from the hot streak has been cited but did not receive the status of being disruptive, the publication has a negative influence on *D_diff* as well. One possible explanation for this effect is that the publication of conventional work might hinder a scientist's ability to engage in disruptive research at the same time. When researchers focus on conventional research, they may not have the time to simultaneously pursue groundbreaking, disruptive research.

Discussion

In this study, by comparing disruptiveness before, during, and after hot streaks, we find that the disruptiveness of scientists' publications is significantly higher during the hot streak than that in their other career stages. Scientists do not only possess high impact during their hot streaks but also exhibit high disruptiveness (at nearly the same time). Although high impact research is not necessarily equivalent to highly disruptive research, our quantitative analyses from a career perspective reveal that periods with highly impactful papers are often the same periods with

Table 6 The influence of a publication with certain characteristics in the fourth model from Table 5 (fictitous scenarious).

	Coefficient	
Feature of a publication	**Before the hot streak**	**During the hot streak**
Without any citations	−0.031	0.029
Has been cited but is not disruptive	0.022	−0.097
Has been cited and is disruptive	−0.933	1.065

highly disruptive papers. The results confirm thus the assumed interplay between both phenomena that we theorized through career mechanisms rooted in cumulative advantage, phases of exploration and exploitation, and paradigm shifts at the beginning of this paper. Although the results do not uncover the concrete mechanisms at play in the interplay, the found coincidence of high impact and disruption especially when the hot streak lasts longer point to their effectiveness. The effectiveness of all career mechanisms requires longer time periods.

Our study also finds that scientists' hot streaks are randomly distributed among careers, which echoes extant analyses (Liu et al., 2018; Sinatra et al., 2016). The production of high impact papers obviously follows the randomness rule, and highly disruptive papers also seem to follow this principle. The empirical existence of the randomness rule supports the applicability of the chance configuration theory (Simonton, 1989) in the context of research on disruptiveness. The random allocation of opportunities and resources in careers may play a significant role in the generation of disruptive outcomes in scientific research.

In our statistical analyses, we test the impact of different aggregation methods (disruption and citation values) on our results and find that the main differences lie in the positive and negative relationships between the length of hot streaks and the disruptiveness of authors' works. However, these variations can be explained by the chance configuration theory (Simonton, 1989). When we measure the disruptiveness of authors' works by selecting their most disruptive papers or by aggregating the disruption of their papers, we find that the longer the hot streak, the greater the disruptiveness. In contrast, when we use other aggregation methods—methods that may be influenced by many low-disruptive papers—the longer the hot streak, the lower the disruptiveness of authors' works appears to be. Although the coefficients have opposite signs, they represent different aspects of the same trend: a longer hot streak corresponds to an increase in the total number of publications an author produces. At the same time, while the most disruptive publications become even more disruptive, the overall average and median disruption scores decline due to the presence of numerous low-disruptive publications.

In this study, we find that the most disruptive years of scientists tend to be published significantly earlier than their most impactful years. The average time for publishing highly disruptive papers is around 14–16 years after scientists start their careers. At this stage, scholars may tend to focus on more disruptive research, despite the high risks that may be associated with this tendency. They may be well-qualified in this phase to have groundbreaking ideas and to undertake risks to pursue them. We assume that, after establishing academic reputation and successful community networks, research of these scientists is more readily accepted and cited by peers than research by less established scientists. This phenomenon has been widely discussed as "preferential attachment" (Fortunato et al., 2018; Jeong et al., 2003). We find that the average time until publishing highly impactful papers is more than 16 years.

The current study suffers from some limitations. First, while we control for time-sensitive biases, e.g., in citation counts, the current empirical results may be biased towards confounding factors such as institutional resources where scientists work and field-specific publishing norms. For example, scientists in resource-intensive research institutions probably have better conditions for hot streaks and disruptiveness compared with scientists in universities without these resources. Second, we select scientists with at least 20-year-long careers and conduct regression analyses based on these scientists. While this filter echoes existing studies on hot streaks (e.g., Liu et al., 2018), those with shorter careers but making prominent contributions to science are ignored. Future research may (additionally) include scientists with shorter careers. Thirdly, we have not explored all possible details of the results, e.g., the standard deviation of mean disruption values during the hot streak may be higher than those mentioned in other studies. While our analysis focuses on disruptive publications and mostly excludes consolidating works, future research could separate disruption and consolidation dimensions to more thoroughly understand the distinct roles these types of work play during a hot streak. This could offer more nuanced insights into the mechanisms at play during the periods of high scientific productivity. Fourthly, we identified the well-known economist John Maynard Keynes (1883–1946) as MAG author ID 1830319489 in our dataset. The publication records assigned to the ID are spanning from 1925 to 2010, which may be inconsistent with the fact that he passed away in 1946. Since eminent scientists can have post-mortem publications, e.g., publications that are based on their comprehensive estate, Keynes' publication set may be valid. However, it is also possible that Microsoft assigned publications from another author to Keynes' author-ID. Such uncertainties are inherent to all bibliometric datasets on the individual level (Tekles and Bornmann, 2020). The detailed manual control of publications that have been assigned to author IDs cannot be achieved in a comprehensive study including several thousand author IDs.

Conclusions

Our study provides insights into the dynamic relationship between research disruptiveness and hot streaks throughout scientists' careers. By leveraging the DCG method and analyzing career data from the MAG dataset, we observe that scientists tend to produce their most disruptive research during their hot streaks phases. This tendency reveals that periods of high citation impact are accompanied by high disruptiveness, particularly when the hot streak extends over a longer period. Furthermore, our analysis demonstrates a distinct temporal pattern in the production of disruptive and impactful papers, with disruptive work generally preceding high impact publications. This phenomenon suggests that scientists' early exploratory phases are crucial for later significant contributions, facilitated by the preferential attachment within the scientific community.

While our findings emphasize the stochastic nature of hot streaks, the observed tendency for disruptive work to emerge earlier in careers invites further exploration. Several non-mutually exclusive factors may contribute to this pattern, such as risk-tolerance dynamics and cognitive flexibility. For instance, early-career researchers, less constrained by established reputational stakes, may pursue high-risk, high-reward ideas that challenge prevailing paradigms. Conversely, senior researchers often face institutional pressures to prioritize incremental progress for funding or publication efficiency. These mechanisms remain speculative but highlight testable hypotheses for future work.

Our findings offer nuanced implications for funding strategies by integrating evidence from our analysis with prior studies on scientific careers and disruptiveness. While our results confirm the randomness of hot streak timing (Liu et al., 2018; Sinatra et al., 2016), they also reveal two actionable patterns: (1) the average time to publishing highly disruptive work (14–16 years after career onset) precedes the peak of citation impact, and (2) higher publication volume increases the likelihood of both entering hot streaks and producing disruptive research. These insights align with recent work by Park et al. (2023) who observed declining disruptiveness across science, but diverge from conventional funding models that prioritize early-career "high-risk" grants or late-career consolidation.

First, the temporal gap between disruptive output (early-mid career) and citation recognition (mid-late career) suggests that funding agencies could optimize support by sustaining risk-tolerant grants across career stages rather than concentrating resources on early-career researchers. For example, mid-career scientists—who our analysis shows are often entering their most disruptive phases—may benefit from flexible funding schemes that allow them to build on exploratory work initiated earlier, echoing Weinberg and Galenson's (2019) findings about theoretical economists. Second, the link between productivity and disruptiveness underscores the value of funding programs that incentivize consistent research output, such as multi-year project grants or institutional support for sustained collaboration networks. This aligns with Liu et al.'s (2018) observation that productivity raises the odds of stochastic success but contrasts with narrow "high-impact" metrics that may prematurely terminate promising lines of inquiry.

Critically, our results challenge the assumption that disruptive work is synonymous with early-career risk-taking. While disruptive papers emerge earlier than highly cited papers, their publication is strongly associated with periods of heightened productivity (hot streaks). These periods are randomly distributed across careers. Our results support a portfolio-based funding strategy: agencies might balance targeted grants for mid-career researchers (who are statistically poised to enter disruptive phases) with universal mechanisms to sustain productivity (e.g., protected research time or reduced administrative burdens). Such an approach coheres with Li et al.'s (2023) findings that disruptive work is well-cited but distinct from "impactful" work, necessitating differentiated evaluation criteria.

By grounding the recommendations in our empirical linkage results between hot streaks, disruptiveness, and productivity and contextualizing the results within existing literature—we provide a framework for funders to mitigate the decline in disruptive science (Park et al., 2023) without overfitting policies to transient career stages. Future work may explore how institutional factors (e.g., team size or resource allocation) moderate these dynamics, but our findings already offer a robust basis for rethinking funding temporality and risk tolerance.

Data availability

MAG is the main dataset that we adopt. MAG is an open-available datasets that can be downloaded from https://www.microsoft.com/en-us/research/project/open-academic-graph/. The most relevant code and datasets of our study are available at https://osf.io/mvfjd/?view_only=aef97e77b15a4ddfb4579d7127fa2dd1. Other datasets generated for our study are available from the corresponding author upon request.

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Notes

1 MAG have formed a six-level discipline scheme with 19 disciplines on Level 0 and 292 disciplines on Level 1. Economics and computer science are labels on Level 0.

2 For example, if one researcher has published four publications with citations of 5, 1, 1, and 1, respectively, while another researcher has published only two publications with citations of 4 and 1, the mean citation count of the first researcher's publications (i.e., $(5+1+1+1)/4=2$) is less than that of the second researcher's publications (i.e., $(4+1)/2=2.5$). The first researcher, despite having a lower mean citation rate, may still be considered better overall due to the higher number of publications. Publications can also be interpreted as impact, since the publications can be seen as being perceived by the community (e.g., by being reviewed by and published in a journal). When using metrics that consider both the number and quality (or impact) of publications the first researcher may score higher, reflecting their broader—publication and citation—influence.

3 Here is a toy example. Suppose that there are two scientists, A and B. Scientist A has published three articles with disruption values equal to 0.81, 0.6, and -0.9, respectively, while scientist B has published two articles with disruption values equal to 0.8 and 0.5. As $0.81>0.8$ and $0.6>0.5$, we tend to believe that A seems to be more disruptive in research than B though A has an additional paper that is not quite disruptive (i.e., $D=-0.9$). Yet, if we calculate DCG with all publications of both scientists, DCG of A equals 0.739 and that of B equals 1.115, which is inconsistent with our common belief. This example shows that, involving negative disruption values when calculating DCG seems inappropriate. We thus exclude all negative disruption values and only keep positive values when calculating DCG.

4 Each paper in MAG has many field labels across the six-level scheme. We choose authors in economics on Level 0 and determine their subfields in economics via their publications' labels on Level 1.

5 The separation of publications into disruptive and consolidating parts show that not only the disruptive, but also the consolidating part is significantly greater during the hot streak than before or after (for the field of economics).

6 We identified the well-known economist John Maynard Keynes as the author with the 82 years starting time in the MAG dataset. We discuss this case as an example for possible limitations of our study in the Conclusions section.

7 This means that for each additional year in the start time of a hot streak, the effect on the dependent variable is −0.002 (0.004 from *start_hot_period* minus 0.006 from *start_year_hot*). Compared to the coefficient for *length_hot*, this effect size differs by about 113 times. When compared to variables with different units, such as the number of publications before (#pub_before) or during (#pub_during) the hot streak, the effect size is about 15 times smaller. Therefore, we conclude that this effect size is relatively small.

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Author contributions

All authors contributed to the conception of the study and the revision of the manuscript. The first author conducted the experiments and drafted the initial version of the paper.

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The authors declare no competing interests.

Ethical approval

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